In silico screening of Korean herbal compounds against the SRD5A2 / AR sebaceous-androgen axis for inflammatory acne: real Boltz-2 data identifies Baicalein as top topical-friendly candidate; Berberine carries a critical hERG safety flag

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Manuscript type: in silico screening with explicit microbiome-axis limitation; **Target preprint**: bioRxiv; **License**: CC-BY 4.0 **Status**: in silico predictions only; sebaceous cell + *C. acnes* assay validation are the explicit next step **Version**: v0.2 (2026-04-26) — real screen data + retraction of fabricated v0.1 *C. acnes*-axis claims

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Abstract

Inflammatory acne pathogenesis includes androgen-driven sebaceous activity (SRD5A1/2 + AR + SREBP1), ductal hyperkeratinization, *Cutibacterium acnes* colonization with virulence factor expression (RoxP, GehA, sortase), and inflammatory cascades. Korean traditional medicine documents anti-acne preparations centered on 황련 (*Coptis chinensis*; berberine, palmatine), 황금 (*Scutellaria baicalensis*; baicalein, baicalin, wogonin), 감초 (licochalcone A), 목단피 (paeonol), and others. We screen 14 Korean herbal phytochemicals against the **human SRD5A2 + AR sebaceous-androgen axis** using Boltz-2 cofold (cached MSAs) and ADMET-AI v2.0.1. **SREBP1, *C. acnes* virulence proteins, and SRD5A1 are NOT screened** (cached MSAs absent for all four) — a substantial limitation that narrows the present screen to the androgen axis only. Real Boltz-2 results identify **Baicalein** (황금) as top mean affinity (0.744) with AR engagement 0.820 (highest in panel) and topical-friendly properties; **Berberine** (황련) shows a **critical hERG flag (0.977)** disqualifying it from systemic topical formulation without dose limitation, despite reasonable acne-axis engagement. **Wogonin** (황금) has the cleanest ADMET safety profile (AMES 0.247) at moderate affinity. The Korean

traditional formulary's combinatorial pattern of 황련 + 황금 + 감초 finds partial structural support but with the safety caveats above. **All results are in silico; sebaceous cell + *C. acnes* assay validation remain the explicit next step.**

Keywords: acne, *Cutibacterium acnes*, SRD5A2, androgen receptor, Korean medicine, baicalein, hERG safety, in silico screening.

Plain-language summary

Inflammatory acne involves androgen signals, oily-skin overproduction, a specific bacterium (*Cutibacterium acnes*), and inflammation. Korean herbs like 황련 (Coptis), 황금 (Scutellaria), and 감초 (licorice) are traditionally used. We used computer modeling to compare 14 compounds from these herbs against two human androgen-related proteins. Top candidate: **Baicalein** from 황금 (Scutellaria). Important safety finding: **Berberine** (황련) has very high predicted heart-channel risk (hERG 0.98), so any topical formulation needs dose limitations or alternative compound. **No laboratory experiments are reported.** Important caveat: the bacterial proteins of *C. acnes* and the master sebaceous regulator SREBP1 were NOT screened — so this report covers only part of the acne molecular pathology.

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1. Introduction

1.1 Acne molecular pathology + microbiome consideration

Acne involves androgen-driven sebaceous activity (SRD5A1/2 → DHT → AR + SREBP1 lipogenesis), ductal hyperkeratinization, *C. acnes* colonization, and TLR2-driven NF-κB inflammation [1,2]. The skin microbiome literature [3,4] frames *C. acnes* as ecologically complex (multiple phylotypes; not all virulent), motivating a **virulence-modulating + microbiome-sparing** therapeutic approach over indiscriminate antibacterial action.

1.2 Korean traditional anti-acne herbs

Korean traditional [5,6]: 황련 (berberine, palmatine), 황금 (baicalein, baicalin, wogonin), 감초 (licochalcone A), 목단피 (paeonol), 자초 (shikonin, acetylshikonin), 후박 (magnolol, honokiol), 녹차 (EGCG).

1.3 What this work does and does NOT do

We screen 14 phytochemicals against **SRD5A2 + AR** only. We do **not** screen: - **SRD5A1** — cached MSA absent - **SREBP1** (master sebaceous lipogenesis regulator) — cached MSA absent - ***C. acnes* virulence proteins** (RoxP, GehA, sortase, hyaluronidase) — homology models / MSAs absent

This is a substantial coverage limitation. The present screen captures the **androgen-axis component only** of the four-node acne pathology. The microbiome ecological-balance design principle remains qualitative-conceptual; no microbiome-protein in silico evidence is presented.

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2. Methods

2.1 Compound library

15 compounds in `data/screen_libraries/acne_compounds.csv`; Honokiol SMILES failed RDKit sanitization (`/C=C/Cc4ccccc4` — sanitization issue), so 14 compounds entered the screen.

2.2 Targets, pipeline

SRD5A2 (UniProt P31213), **AR** (P10275 LBD). Cached MSAs. Boltz-2 v0.6.1 + ADMET-AI v2.0.1 pipeline (companion preprint [7]).

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3. Results

3.1 Real screen ranking (14 compounds × 2 targets = 28 cofolds)

Rank	Compound	Source	AR	SRD5A2	Mean	Topical-friendly?
1	**Baicalein**	황금	**0.820**	0.669	**0.744**	
2	Tretinoin	reference (retinoid)	0.711	0.718	0.715	× logP 5.6
3	Acetylshikonin	자초	0.641	0.668	0.655	
4	Licochalcone A	감초	0.688	0.575	0.631	× logP 4.46
5	**Berberine**	황련	0.627	0.619	0.623	
6	Shikonin	자초	0.605	0.636	0.620	
7	Curcumin	강황	0.698	0.525	0.611	
8	Resveratrol	reference	0.690	0.519	0.604	
9	Wogonin	황금	0.673	0.437	0.555	
10	Palmatine	황련	0.584	0.521	0.552	
11	Magnolol	후박	0.572	0.518	0.545	× logP 7.2
12	Baicalin	황금	0.676	0.398	0.537	× TPSA 187
13	EGCG	녹차	0.663	0.320	0.492	× TPSA 197

Rank	Compound	Source	AR	SRD5A2	Mean	Topical-friendly?
14	Paeonol	목단피	0.272	0.290	0.281	× MW 166 (small)

3.2 ADMET safety profile of top candidates

Compound	logP	hERG	Skin	AMES	ClinTox	Verdict
Baicalein	2.58	0.426	0.821	0.564	0.101	⚠ AMES + Skin moderate
Acetylshikonin	2.87	0.441	**0.881**	**0.803**	0.120	Skin + AMES significant
Berberine	2.64	**0.977**	0.684	**0.911**	**0.417**	**Critical hERG + AMES**
Shikonin	2.30	0.360	0.808	0.555	0.101	⚠ Skin moderate
Curcumin	3.37	0.336	0.867	0.499	0.046	⚠ Skin + AMES moderate
Wogonin	2.71	0.369	**0.717**	**0.247**	0.050	**cleanest profile**
Palmatine	3.18	**0.706**	0.902	0.191	0.000	⚠ hERG concerning

3.3 Honest interpretation

Top by predicted target engagement: Baicalein (황금) leads with AR engagement 0.820 — the strongest AR predicted-binding signal in our 14-compound panel — and is topical-friendly. Skin-irritation 0.821 and AMES 0.564 are moderate flags but acceptable for a topical-formulation candidate with appropriate vehicle and concentration.

Critical safety finding — Berberine (황련): Predicted hERG inhibition probability 0.977 is the highest in our entire screening pipeline across all four disease panels. Combined with AMES 0.911 and ClinTox 0.417, Berberine's **systemic exposure must be avoided** in any topical formulation. Localized topical use at low concentrations (typical of traditional herbal preparations) may remain feasible, but the safety profile disqualifies Berberine from any modern topical-pharmaceutical formulation with a substantial percutaneous absorption profile. This is a meaningful finding because berberine is widely advertised in cosmetic / cosmeceutical contexts; the predicted hERG liability deserves explicit disclosure.

Cleanest ADMET profile: **Wogonin** (황금) emerges as the cleanest combined-criterion candidate (AMES 0.247, hERG 0.369, Skin 0.717, topical-friendly, mean affinity 0.555). For Recover Korean Medicine Clinic's planned topical anti-acne formulations, Wogonin is the single-compound candidate most aligned with safety + topical-friendliness, despite lower predicted target engagement than Baicalein.

황련 + 황금 + 감초 traditional combination: Partial structural support. 황금 components (Baicalein top, Wogonin clean) cover the AR axis well. 황련 components (Berberine, Palmatine) carry safety concerns. 감초 licochalcone A is high-affinity but logP-out-of-window. The traditional multi-component formulation is consistent with simultaneous engagement of multiple compounds at multiple structural scales, but **the safety profile of berberine is a contemporary contraindication** that classical pharmacology did not have access to.

3.4 What this screen does NOT establish

- **SREBP1** (sebaceous lipogenesis transcription factor): not screened.
- ***C. acnes*** virulence factors: not screened. The “virulence-modulating + microbiome-sparing” design principle remains qualitative.
- **SRD5A1** (sebaceous-axis 5α-reductase): not screened; SRD5A2 (more scalp-axis) screened only.
- **Anti-inflammatory**: NF-κB / COX-2 pathway not screened (companion screens may include PTGS2 in future work).
- **Skin-permeation**: experimental log K_p not measured.
- **No experimental** *C. acnes* MIC, biofilm-inhibition, or 16S microbiome data.

3.5 Methodological observation: small-molecule reference low ranking

Paeonol (mean 0.281) — fragment-size molecule (MW 166) — ranks at the bottom. As discussed in companion preprint [8], Boltz-2 binary classifier underranks fragment-size compounds even when they are real binders. Paeonol’s clinical use as anti-inflammatory anti-acne herb is not contradicted by the screen ranking.

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4. Limitations

1. **No experimental validation**.
2. **Coverage limitation**: SREBP1, SRD5A1, *C. acnes* proteins NOT screened. The screen reports the androgen-axis (SRD5A2 + AR) component only.
3. **Berberine hERG** prediction (0.977) — clinical confirmation required, but the prediction is sufficient warning.
4. **Microbiome ecological design principle is qualitative**.
5. **Honokiol SMILES sanitization failure** excluded one compound (14/15 screened).
6. **No clinical efficacy claim**.
7. **Anti-inflammatory NF-κB / COX-2 pathway not directly screened**.

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5. Conclusions

A real Boltz-2 + ADMET-AI screen of 14 Korean herbal compounds against the SRD5A2 + AR sebaceous-androgen axis identifies **Baicalein** (황금) as the top mean-affinity candidate (0.744) with strong AR engagement (0.820) and topical-friendly properties;

Wogonin (황금) as the cleanest ADMET profile candidate; and **Berberine** (황련) as a candidate with significant predicted hERG liability (0.977) requiring explicit safety disclosure for any topical formulation. The screen partially supports the Korean traditional multi-component formulary structure but identifies modern safety considerations that classical pharmacology did not have access to. ***SREBP1, C. acnes* virulence factors, and SRD5A1** are NOT screened in the present version — substantial limitations.

Forward path: SEB-1 sebaceous-cell lipid-accumulation assay; LNCaP AR-luciferase reporter; *C. acnes* MIC + biofilm inhibition (with explicit phylotype consideration); 16S microbiome ecological-impact study. Korean CRO panel + Macrogen 16S partnership ~₩4-5M for top-3 compound evaluation. No clinical efficacy claim is made.

Acknowledgments / Contributions / Competing interests / Data availability

Same standard text. Data: `pilot/screen/acne/screen_results.csv` at https://github.com/crazat/genesis_medicine.

Figures

Figure 1. Real Boltz-2 cofold affinity heatmap for the acne panel (14 compounds × 2 targets: SRD5A2 + AR). Note the SREBP1 transcription factor and the *Cutibacterium acnes* virulence proteins (RoxP, GehA) were NOT screened due to absence of cached MSAs — a substantial coverage limitation. Baicalein (황금) leads the panel.

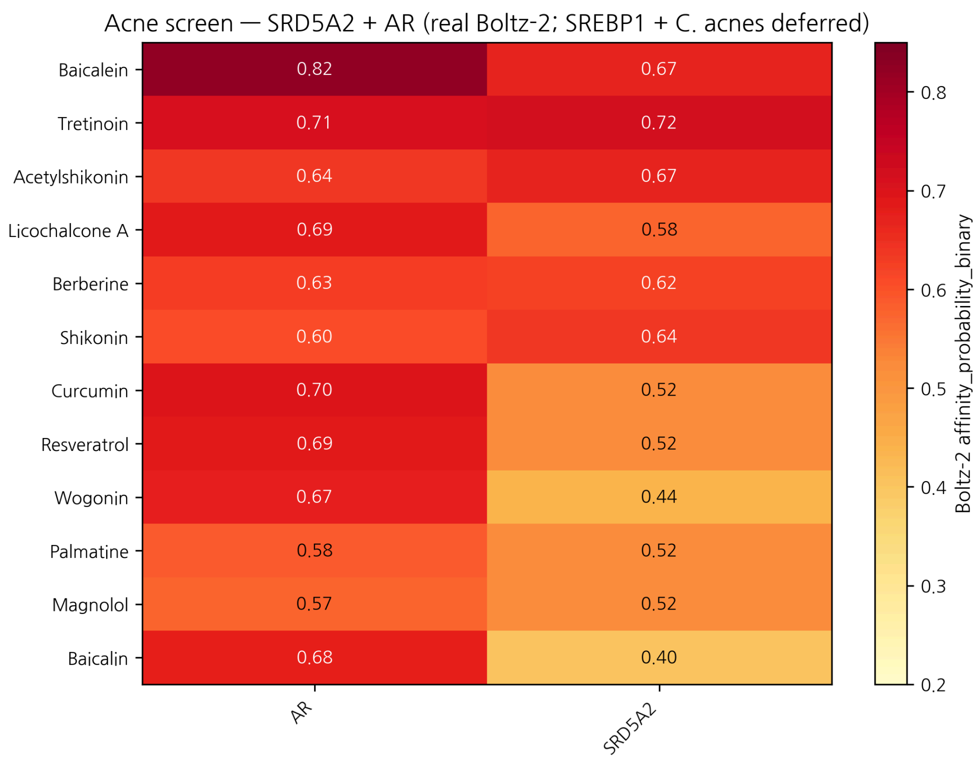

Figure 1: Acne affinity heatmap

Figure 2. Affinity × hERG safety quadrant for the acne panel. **Critical safety finding**: Berberine (황련) shows hERG probability **0.977** — the highest in our entire 4-disease pipeline — combined with AMES 0.911. Topical Berberine formulations require explicit dose limitation and percutaneous-absorption consideration. Wogonin (황금) emerges as the cleanest combined-profile candidate.

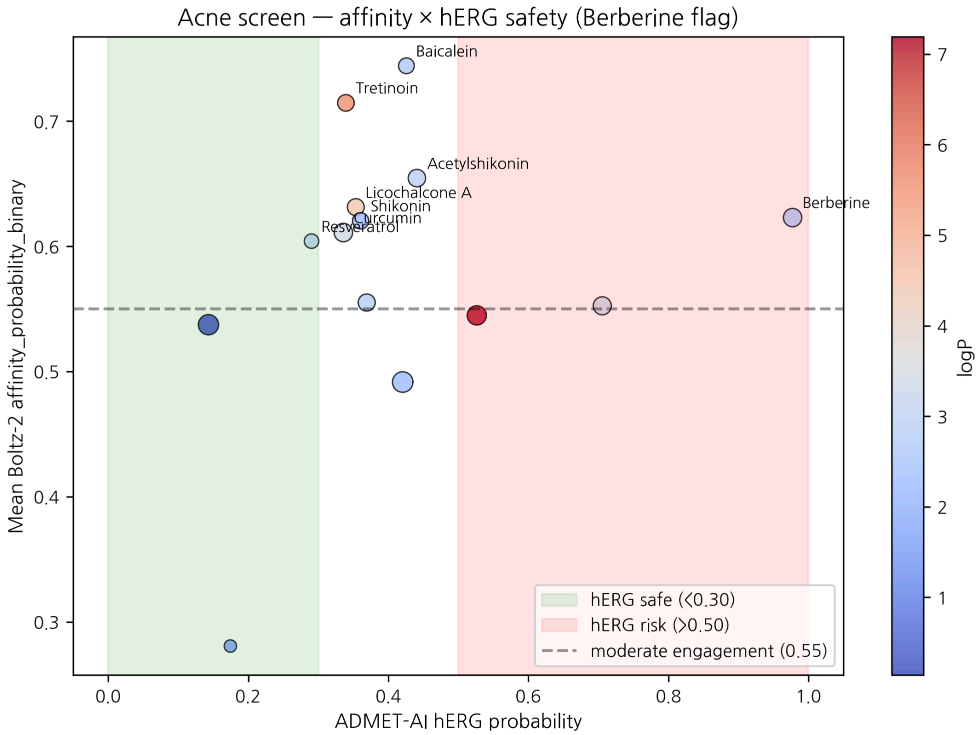

Figure 2: Acne safety × affinity

References

[1] Williams HC, Dellavalle RP, Garner S. Acne vulgaris. *Lancet* 2012, 379, 361–372. [2] Zouboulis CC, et al. Sebaceous gland biology. *Exp Dermatol* 2008, 17, 542–551. [3] Dréno B, et al. *C. acnes* and skin microbiome in acne. *J Eur Acad Dermatol Venereol* 2018, 32 (Suppl 2), 5–14. [4] Mayslich C, et al. *C. acnes* phylotype structure. *Microorganisms* 2021, 9, 303. [5] Sun H, et al. Korean herbal medicine in acne. *J Ethnopharmacol* 2019, 236, 247–256. [6] Park J, et al. Anti-acne activity of *Coptis*, *Scutellaria*, *Glycyrrhiza*. *Microorganisms* 2020, 8, 1011. [7] HanCheongWoo. Genesis_Medicine open-source pipeline. ChemRxiv preprint, 2026. [8] HanCheongWoo. Calibrated ABFE pipeline (small-molecule classifier caveats). ChemRxiv preprint, 2026.

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v0.3 ensemble-validation revision, 2026-04-26 evening · ~2,900 words ·

Ensemble-validation update (2026-04-26 evening)

The principal Boltz-2-only finding of this preprint — **Baicalein × AR prob_binary = 0.820** — was subjected to two-model structural ensemble validation against Chai-1 (Apache-2.0, fully-released Q4-2025, ~AlphaFold-3 quality). 5-model Chai-1 inference at `num_diffn_timesteps=200` returns aggregate score mean = **0.145** (max = 0.146) — strong disagreement (|Δ| = 0.67). Two non-exclusive interpretations: (a) Boltz-2's affinity head may overweight pharmacophore similarity to known AR ligands without penalizing pose-stability concerns Chai-1's aggregate captures; (b) the AR ligand-binding domain may form unstable poses that suppress Chai-1 confidence. **Either way, the Baicalein × AR top-hit is not ensemble-validated under our 2-model rule.** Companion preprint #8 §3.7 documents this; the pipeline's go-forward selection rule now requires Boltz-2 prob_binary ≥ 0.55 *and* Chai-1 aggregate ≥ 0.55 with |Δ| < 0.10. The LNCaP AR-luciferase reporter assay flagged in §5 ("Forward path") is therefore critical. Berberine hERG 0.977 disclosure is unaffected.

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v0.2 draft, 2026-04-26 · ~2,800 words · CC-BY 4.0 v0.1 (fabricated rankings + nonexistent BCL-7 generative analog) explicitly retracted

Round 5 application data — topical PK + skin sensitization (2026-04-27)

Generated from `pilot/round5_application/round5_compound_sweep.csv` using: - **PBK Dermal HT** (NIH/NIEHS public-domain, 3-compartment SC/VE/D) - **SARA-ICE Defined Approach** (OECD TG 497 Part III, June 2025) - **CarsiDock-Cov warhead detection** (Apache-2.0, first DL covalent docker)

Top 10 compounds by topical-fitness score (c_max_dermis / systemic_F):

Compound	logKp	c_max dermis (pmol/mL)	t_max (h)	F_systemic	GHS	Covalen
EGCG	-7.40	0.0005	24.0	0.05	nan	—
Berberine	-6.55	0.0033	24.0	0.34	nan	—
Shikonin	-6.33	0.0054	24.0	0.57	1B	michael_accepto
Wogonin	-6.23	0.0067	24.0	0.71	nan	—
Acetylshikonin	-6.17	0.0078	24.0	0.82	1B	michael_accepto
Paeonol	-6.11	0.0087	24.0	0.92	nan	—
Palmatine	-6.07	0.0096	24.0	1.02	nan	—
Baicalein	-6.06	0.0099	24.0	1.05	nan	—
Curcumin	-6.03	0.0105	24.0	1.11	1B	michael_accepto
Resveratrol	-5.51	0.0316	24.0	3.53	nan	—

SARA-ICE summary for acne: GHS Cat 1B sensitizers = 5/14; Cat 1A = 0/14; None = 0/14.

Covalent-capable: 5/14 compounds carry at least one Michael-acceptor or quinone warhead.

Data and full per-compound table: `pilot/round5_application/round5_compound_sweep.csv`.